# Quantum error Correction

4/9/2026

- Nielsen + Chuang Chapter 10 } textbooks
- Westmoreland + Schumacher

Think about quantum channels as maps

- $\rho \to \rho' = \mathcal{E}(\rho)$

- relaxation or decoherence of the system that negatively change the physical properties of the system.
- Logical information is encoded in the physical state + evolution of a system
  - noise affects the physical state ⇒ it creates logical errors in encoded information.

eg
- binary information encoded in electronics ⇒ cosmic rays
- rosetta stone ⇒ weathering / theft / change of language
- ink on paper ⇒ dogs
- braille
- tonings ⇒ forget

Dealing with errors in physical representation of information for a long time

⇒ Redundancy is the answer

≈ backups

navigation problem ⇒ 3 clocks.

timing inaccuracy ⇒ "majority" vote or average.

QEC is about adding redundancy to encode of quantum information:

- Use multiple physical systems to redundantly encode logical information.

Prototypical classical error correction is called the "repetition" code: repeat a message multiple times

encode ∈ of physical system

$0_L \longrightarrow 000$

$1_L \longrightarrow 111$

9/9/2026 QEC

What is the relative "strength" of errors in classical vs. quantum systems.

"Bit Error Rate" (BER) - rate at which a given physical bit of information is changed eg. Bit-flip 0 ⇔ 1 or erasure

Clock speed - rate at which gates are performed eg GHz on a laptop

| | Classical | Quantum |
|---|---|---|
| BER | Typical hard disk<br>~ 1 TB of data = $10^{13}$ bits<br>1 bit is a small magnetic domain<br>estimate the BER for a hard disk<br>• 1 bit per year / hard disk is ok<br>So BER = $\frac{1 \text{ bit error/year/disk}}{10^{13} \text{ bits/disk} \times 10^{7} \text{ sec/year}}$<br>$= 10^{-20}$ bit errors/sec | A very good qubit (superconducting) has a typical lifetime of ~ 1 ms<br>$\Rightarrow$ BER = $10^{3}$ errors/sec |
| | $10^{-23}$ | |
| Clock speed | ~ GHz ~ $10^{9}$ ops/sec | 1 - 10 MHz<br>$\Rightarrow 10^{7}$ op/sec |
| | $10^{2}$ | |
| BER/op | $10^{-29}$ errors/op. | $10^{-4}$ errors/op. |
| | $10^{-25}$ | |

So quantum computing tech is orders of magnitude worse than digital at present

- But we hope that q. computers will be useful ⇒ so we try to fix the high rate to get a practical q.c.

⇒ Quantum error correction

Quantum errors are caused by quantum noise → Quantum channels.

A "standard" error model: depolarising channel

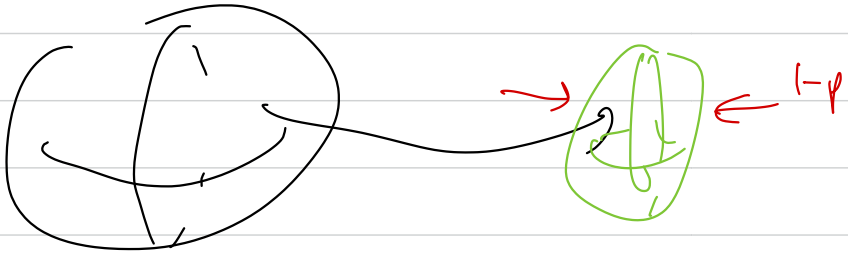
$$\rho \to \rho' := \mathcal{E}_{dep}(\rho) = (1-p)\frac{\mathbb{1}}{2}\rho\mathbb{1} + \frac{p}{3}\left(X\rho X^\dagger + Y\rho Y^\dagger + Z\rho Z^\dagger\right)$$

$$\mathcal{E}_k = \left\{\sqrt{1-p}\,\mathbb{1}, \sqrt{\tfrac{p}{3}}X, \sqrt{\tfrac{p}{3}}Y, \sqrt{\tfrac{p}{3}}Z\right\}$$

(check $\sum_k \mathcal{E}_k^\dagger \mathcal{E}_k = \mathbb{1}$

Symmetric in $X, Y$ & $Z$

⇒ Bloch sphere shrinking uniformly



Repeated applications of the depolarizing channel

$$\rho \to \mathcal{E}(\rho)^n \xrightarrow[n\to\infty]{} \mathbb{1}/2$$

Conceptually think of a random process

| | | | "Name" |
|---|---|---|---|
| classical processes | $1-p$: $\lvert 4\rangle$ | $\mathbb{1}\rho\mathbb{1}$ | identity / no error |
| | $p$: $\lvert 4\rangle \to X\lvert 4\rangle$ | $\rho \to X\rho X^\dagger$ | bit flip $0 \leftrightarrow 1$ |
| quantum processes | $p/3$: $\lvert 4\rangle \to Y\lvert 4\rangle$ | $Y\rho Y^\dagger$ | bit + phase flip $Y = iXZ$ |
| | $p/3$: $\lvert 4\rangle \to Z\lvert 4\rangle$ | $Z\rho Z^\dagger$ | phase flip $\lvert 0\rangle \leftrightarrow \lvert 0\rangle$, $\lvert 1\rangle \to -\lvert 1\rangle$, $\lvert +\rangle \leftrightarrow \lvert -\rangle$ |

We don't know which random effect occured.

Depolarizing Channel contains classical + quantum error processes so it's a good hardware indep. model of quantum errors.

Minimal goal of QEC: correct + protect against the depolarizing channel

Classical / Digital error correction is a 2-state bit.

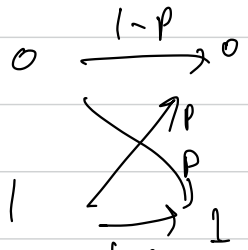
- Bit flip errors $0 \leftrightarrow 1$
- erasure

Representing data (information) is a physical encoding

Bit string: $\underbrace{0,1,0,1,\ldots}_{n \text{ - physical systems}} \in \{0,1\}^{\otimes n}$

each physical system undergoes a bit flip error

Binary Symmetric Channel




→ time (storage + retrieval)

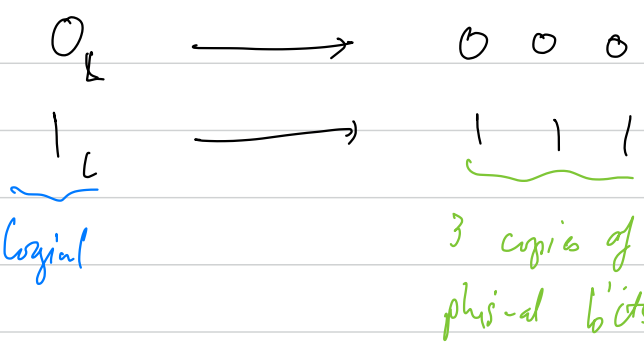
→ space (communication)

is the prototype classical error process

Task: store or send logical information reliably even with bit-flip errors.

reliably: if we send 1 logical bit per physical system, we can send $n \sim \frac{1}{p}$ bits before a likely failure

- we want $n \gg \frac{1}{p}$ bits reliably
- use redundancy to achieve this. "encode" logical data in multiple physical systems




- "3-bit repetition code" [$(2n+1)$-bit repetition code]
- assume bit flip errors happen independently + identically distributed (iid) on physical bits
- is not perfect: 3 errors could happen ⇒ corrupt the logical information.
- 3-bit rep. code can reliably correct 1 error
- " " " " " detect 2 errors, but not correct

3-bit rep code is designed to correct error

~ "distance" $d$ of the code is smallest # of undetectable errors

$d = 3$ for 3-bit rep code.

- can detect $t = d - 1$ errors
- can correct $\left\lfloor \frac{d-1}{2} \right\rfloor$ errors

notation $[n, k, d] = [3, 1, 3]$ for a 3-bit rep code.

$n$: # physical bits
$k$: # logical bits
$d$: code distance

Scenario: ~~encode~~ $0_L = 000$ send

↓ send + bit flip on $b_3$

001 recieve

↓ "decode"

$000 = 0_L$ ✓

alternatively $1_L = 111$ send

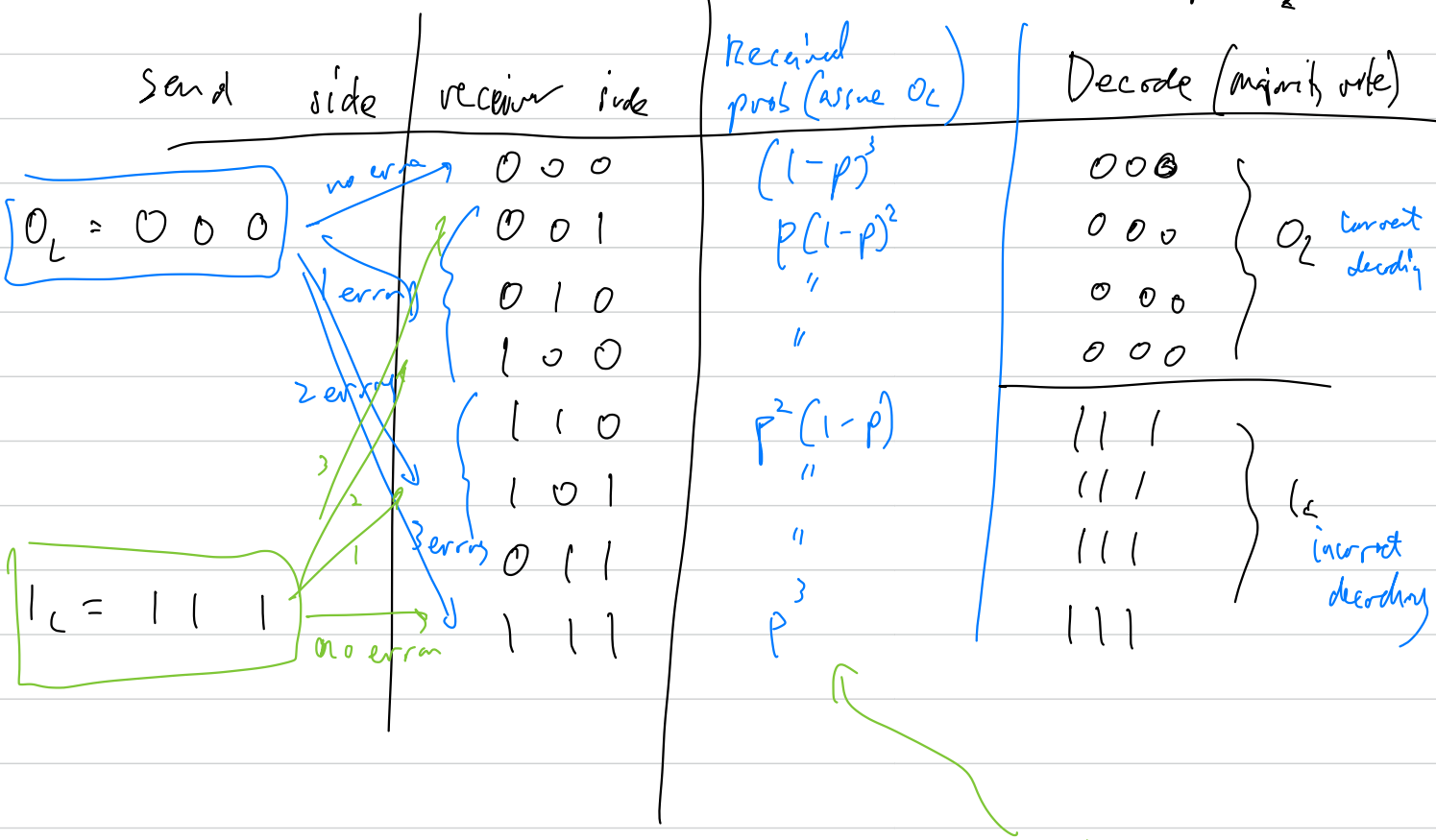
↓ bit flip $b_1$ + $b_2$

001 receive.

↓ decode

$000 = 0_L$ ✗

Formalise this for b.s.c. channel with bit-flip prob $p < \frac{1}{2}$




if $1_L$ was sent, just reverse the probs

$$P(\text{successful decoding} \mid 0_L \text{ sent}) = (1-p)^3 + 3p(1-p)^2 = 1 - 3p^2 + 2p^3$$

$$P(\text{unsuccessful} \mid 0_L \text{ sent}) = p^3 + 3p^2(1-p) = 3p^2 - 2p^3$$

So if $p \ll \frac{1}{2}$ then $p^2 \ll p$

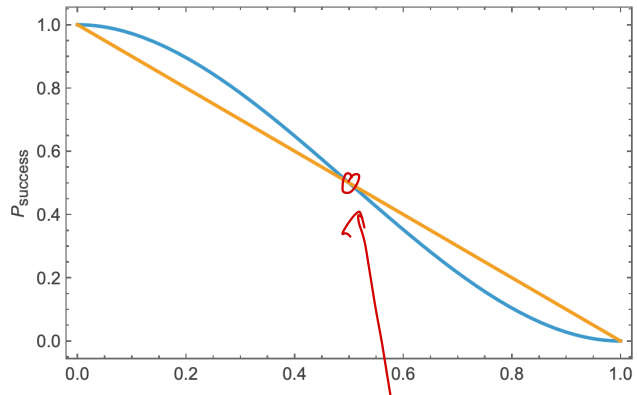
So $P(\text{success} \mid \text{encoding}) \simeq 1 - 3p^2 > \underbrace{1-p}_{\text{success rate with no encoding}} = P(\text{success} \mid \text{no encoding})$

$\Rightarrow P(\text{fail} \mid \text{encoding}) \sim 3p^2 \ll p = P(\text{fail} \mid \text{no encoding})$

combinatorial factor — redundancy

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In[•]:= Plot[{(1 - p)^3 + 3 p (1 - p)^2, 1 - p}, {p, 0, 1}, Frame → True, FrameLabel → {p, P_success}]
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Out[•]=




encoding is better than not encoding

threshold where encoding doesn't make a difference

Codes typically have an error threshold $p^*$

so that if $p < p^*$, encoding helps

- At high error rates, more places for errors to occur $\Rightarrow$ encoding degrades

aside: really the receiver is trying to determine

$$\max \begin{cases} P(0_L \text{ sent} \mid \text{data string } \underline{d} \text{ received}) \\ P(1_L \text{ sent} \mid \text{"} \quad \underline{d}) \end{cases}$$

receiver computes

$$\begin{matrix} P(\text{data } \underline{d} \mid \text{sent } x_L = 0_L) \\ P(\underline{d} \mid \text{sent } x_L = 1_L) \end{matrix}$$

Bayes' rule : $P(A|B) = P(B|A)\,\frac{P(A)}{P(B)}$

eg here $P(0_L \text{ sent} \mid \underline{d}) = P(\underline{d} \mid 0_L \text{ sent})\,\frac{P(0_L \text{ sent})}{P(\underline{d} \text{ was received})}$

$P(1_L \text{ sent} \mid \underline{d}) = P(\underline{d} \mid 1_L \text{ sent})\,\frac{P(1_L \text{ sent})}{P(\underline{d} \text{ was received})}$

from receivers point of view $P(0_L \text{ sent}) = P(1_L \text{ sent}) = \frac{1}{2}$

so relative likelihood

$$\frac{P(0_L \text{ sent} \mid \underline{d})}{P(1_L \text{ sent} \mid \underline{d})} = \frac{P(\underline{d} \mid 0_L \text{ sent})}{P(\underline{d} \mid 1_L \text{ sent})}$$

eg if $\underline{d} = 001$

$$LHS = \frac{p(1-p)^2}{p^2(1-p)} = \frac{1-p}{p}$$

if $p << \frac{1}{2}$ then $P(0_L \text{ sent} \mid \underline{d}) >> P(1_L \text{ sent} \mid \text{data})$

$0_L$ is the most likely sent bit.

→ Maximum likelihood decoding

# Quantum repetition Code

In QEC we have unknown quantum information

$$|\psi_{(1)}\rangle = \alpha|0\rangle + \beta|1\rangle \qquad \alpha \text{ & } \beta \text{ are unknown}$$

$$|\psi_{(2)}\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

$$\vdots$$

We want to reliably store or transmit unknown quantum states

**Guess A** (logical states)

$$|\psi_L\rangle \rightarrow |\psi\rangle_a \otimes |\psi\rangle_b \otimes |\psi_c\rangle$$

($a$, $b$, $c$: physical states)

**Why** no cloning $\Rightarrow$ can't copy quantum states

There is no unitary operation that achieves

$$U|\psi\rangle \otimes |0\rangle = |\psi\rangle \otimes |\psi\rangle$$

- Brute force calc
- Colligal statement :
  - $U$ is a linear map
  - LHS is linear in $\alpha$ & $\beta$
  - RHS is quadratic in $\alpha$ & $\beta$

Guess B — encode in "code words" 000 + 111

$$|0_L\rangle = |000\rangle = |0\rangle_a \otimes |0\rangle_b \otimes |0\rangle_c$$

$$|1_L\rangle = |111\rangle = |1\rangle_a \otimes |1\rangle_b \otimes |1\rangle_c$$

different physical objects

Quantum state is encoded as

$$|\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle = \alpha|000\rangle_{abc} + \beta|111\rangle_{abc}$$

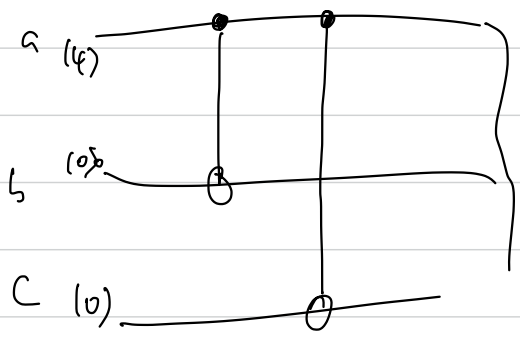
is this compatible with Q.M.

is there a unitary that encodes $|\psi\rangle$ into an encoded state of 3 physical qubits.

$$U\ |\psi\rangle_a \otimes |0\rangle_b \otimes |0\rangle_c = |\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$$

$(\alpha|000\rangle + \beta|100\rangle)$

$U =$




$\alpha|000\rangle + \beta|111\rangle = |\psi_L\rangle$

$$U_{enc} = CNOT_{12} \cdot CNOT_{13}$$

**Problem**: how to detect + decode errors.

we don't see this

$|\psi\rangle \otimes |0\rangle \otimes |0\rangle \xrightarrow{\text{encode}} |\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle \xrightarrow[X_1]{\text{bit flip error}} |\psi'\rangle = X_1|\psi_L\rangle = \alpha|100\rangle + \beta|011\rangle$

"code space" $C_1 \Rightarrow \dim 2$

$|\psi'\rangle \in C_1^{\perp} \ (\dim 4)$

$\langle \psi' | \psi_L \rangle = 0$

$|\psi_L\rangle \xrightarrow{\mathcal{E}} |\psi'\rangle = \mathcal{E}|\psi_L\rangle \xrightarrow[\text{Correction } c]{\text{decode?}} c|\psi'\rangle = c\mathcal{E}|\psi_L\rangle$

if $c\mathcal{E}|\psi_L\rangle = |\psi_L\rangle$ then we have successfully decoded.

Decoding problem: work out the correction operation $c$

- In general we don't know if or where an error occurred
- We need to extract some information from $|\psi'\rangle = \mathcal{E}|\psi_L\rangle$ that will enable a good guess as to the error

Q: what does the receiver do to find the error?

**Guess 1** Measure each physical qubit (in the computational basis) to get a received bit string

Measure $Z_1$, $Z_2$ & $Z_3$ on each physical qubit.

Measure $Z_1, Z_2, Z_3$

$|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle \xrightarrow{\mathcal{E}=X_1} |\psi'\rangle = \mathcal{E}|\psi_L\rangle = \alpha|100\rangle + \beta|011\rangle$

"1", "0", "0" — prob $|\alpha|^2$ — $|100\rangle \to |000\rangle$ — correct $C = X_1$

"0", "1", "1" — prob $|\beta|^2$ — $|011\rangle \to |111\rangle$

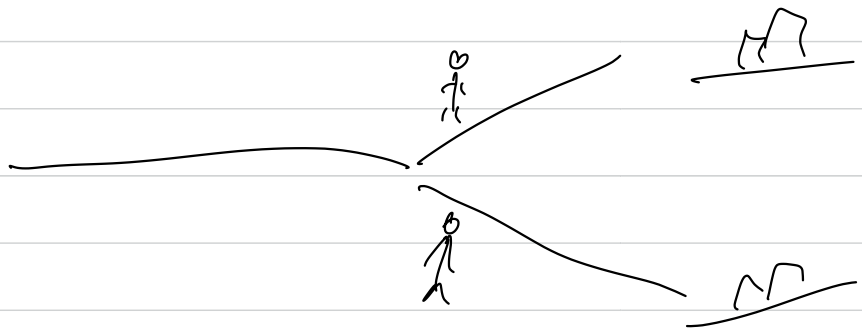
outcomes

$$C.M.\,\mathcal{E}|\psi_L\rangle = \begin{cases} |000\rangle & \text{with prob } |\alpha|^2 \\ |111\rangle & \text{" " } |\beta|^2 \end{cases}$$

$$\neq |\psi_L\rangle$$

How do we avoid the problem that the measurement inherently disrupts the logical qubit?

Answer use "Parity Checks"



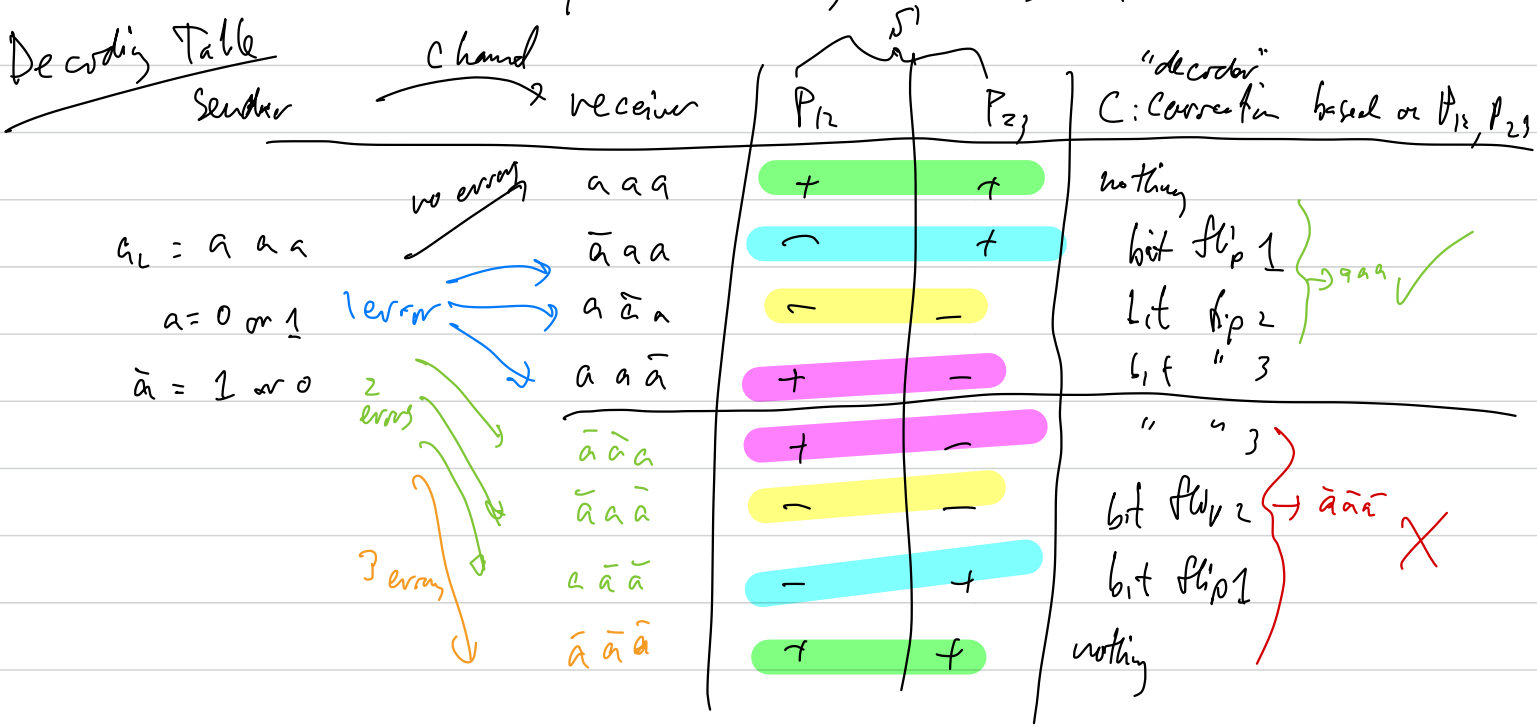
Parity Check : classical case

| | $b_1$ | $b_2$ | $b_3$ | $P_{12}$ | $P_{23}$ |
|---|---|---|---|---|---|
| subject to bit flip on bit 2 | 0 | 0 | 0 | −1 | +1 |
| | 0 | 1 | 1 | −1 | +1 |

After a bit flip on 1, $b_1 + b_2$ disagree with each other
$b_2 \vee b_3$ agree "

$$P_{ij} : \begin{cases} +1 & \text{if } b_i = b_j \\ -1 & \text{if } b_i \neq b_j \end{cases} = (-1)^{b_i + b_j}$$

Enumerate the possible Parity checks for 3 bits:




The decoding table tells us what correction action to take based on the parity check bits.

- "Look-up" table decoder: enumerate all parity check configurations & the corrections.
- "Syndrome" $S'$ is the vector of parity check bits

$$S' = \{P_{12}, P_{23}\}$$

- we use the syndrome to compute the correction

$$C = C(S')$$

- For the repetition code the outcome of this decoder is the same as for the majority vote decoder.

$\Rightarrow$ Parity Check data (/ syndrome) is enough to decode errors

Aside: • what about $P_{13} = (-1)^{b_1 + b_3}$

$$= (-1)^{b_1 + b_2 + b_2 + b_3}$$ ($b_2 + b_2$ even)

$$= (-1)^{b_1 + b_2} (-1)^{b_2 + b_3}$$

$$= P_{12} P_{23}$$

So $P_{13}$ is determined by $P_{12}$ & $P_{23}$: $P_{13}$ is "generated" by $P_{12}$ & $P_{23}$

• # logical bits ($k$) = # physical bits ($n$) − # independent parity checks

(# unconstrained degrees of freedom; total # of degrees of freedom in $n$ bits; # of constraints for valid code words)

for 3-bit rep code: $1 = 3 - 2$ ✓

• valid code words have $+1$ for all parity check bits.

# indep. parity checks $= n - k$

$= [n, k, ?]$ code

# Quantum Repetition code

- We need a "quantum" version of parity check bits

  → measurement of an observable that reveals parity data

- in quantum mechanics an observable is a Hermitian operator

Quantum Parity check for 2 qubits, $\hat{P}_{12}$ has e/vals $\pm 1$ for computational basis state

$\hat{P}_{12}$ should satisfy

$$\begin{cases} \hat{P}_{12}|00\rangle = (+1)|00\rangle \\ \hat{P}_{12}|01\rangle = (-1)|01\rangle \\ \hat{P}_{12}|10\rangle = (-1)|10\rangle \\ \hat{P}_{12}|11\rangle = (+1)|11\rangle \end{cases}$$

$$\hat{P}_{12} = \begin{bmatrix} +1 & & & \\ & -1 & & \\ & & -1 & \\ & & & +1 \end{bmatrix}_{4\times 4}$$

(eg $\langle 00|P_{12}|11\rangle = \langle 00|((+1)|11\rangle) = \langle 00|11\rangle = 0$)

$\hat{P}_{12} = Z_1 \otimes Z_2$

$\hat{P}_{12}^\dagger = (Z_1 \otimes Z_2)^\dagger = Z_1^\dagger \otimes Z_2^\dagger = Z_1 \otimes Z_2 = \hat{P}_{12}$ is Hermitian $\Rightarrow$ observable.

$\hat{P}_{23} = Z_2 \otimes Z_3$ by analogy, check $\hat{P}_{23}$ behaves correctly in systems 2+3

Check $\hat{P}_{13} = Z_1 \otimes Z_3 = (Z_1 \otimes Z_2)(Z_2 \otimes Z_3)$ since $Z_2^2 = \mathbb{1}$

$= \hat{P}_{12} \otimes \hat{P}_{23}$

slightly more formally:

$$\begin{cases} \hat{P}_{12} = Z_1 \otimes Z_2 \otimes \mathbb{1}_3 = Z_1 Z_2 \mathbb{1}_3 \\ \hat{P}_{23} = \mathbb{1}_1 \otimes Z_2 \otimes Z_3 = \mathbb{1}_1 Z_2 Z_3 \end{cases}$$

$$\hat{P}_{12} = (Z_1 \otimes Z_2 \otimes \mathbb{1}_3)(\mathbb{1}_1 \otimes Z_2 \otimes Z_3)$$

$$= (Z_1 \mathbb{1}_1) \otimes (Z_2 Z_2) \otimes (\mathbb{1}_3 Z_3)$$

↑ ordinary matrix multiplication of 2×2 matrices

$$= Z_1 \otimes \mathbb{1}_2 \otimes Z_3$$

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So we have now 2 Hermitian observables

$$\hat{P}_{12} = Z_1 \otimes Z_2$$
$$\hat{P}_{23} = Z_2 \otimes Z_3$$

That each reveal a parity check bit when measured.

Back to the error-prone, state

$$|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle \xrightarrow{\varepsilon = X_1} |\psi'\rangle = \sum_i |\psi_i\rangle$$
$$= \alpha|100\rangle + \beta|011\rangle$$

what do we get if we measure $|\psi'\rangle$ with $\hat{P}_{12}$ ?

– is $|\psi'\rangle$ an e/state of $\hat{P}_{12}$ ?

$$\hat{P}_{12}|\psi'\rangle = Z_1 \otimes Z_2 \left(\alpha|100\rangle + \beta|011\rangle\right)$$

$$= -\alpha|100\rangle - \beta|011\rangle$$

$$= -|\psi'\rangle$$

So measuring $|\psi'\rangle$ with $\hat{P}_{12}$ gives an outcome "$-1$"

- How about $\hat{P}_{23}|\psi'\rangle = Z_2 \otimes Z_3 \left(\alpha|100\rangle + \beta|011\rangle\right)$

$$= \alpha|100\rangle + \beta|011\rangle = (+1)|\psi'\rangle$$

So we get e/vals $\{-1, +1\}$

– Next week : Compatible observables

16/9/2026

- We need to measure both parity check operators : "stabilizers" simultaneously without disrupting the quantum state
  ⇒ simultaneously diagonalizable

⇒ they must commute

check $\left[\hat{P}_{12}, \hat{P}_{23}\right] = \left[Z_1 Z_2 \mathbb{1}_3, \mathbb{1}_1 \otimes Z_2 \otimes Z_3\right]$

$= (Z_1 \mathbb{1}_1) \otimes (Z_2 Z_2) \otimes (\mathbb{1}_3 Z_3) - (\mathbb{1}_1 Z_1) \otimes (Z_2 Z_2) \otimes (Z_3 \mathbb{1}_3)$

$= 0$ ✓

So in the ex we have syndrome data $\{-1, +1\}$

- which is consistent with either $\mathcal{E} = X_1$ or $\mathcal{E} = X_2 X_3$

— Bayesian inference we guess $\mathcal{E} = X_1$ as more likely

⇒ $G^{\dagger} = X_1$

$|\psi_1\rangle = \alpha|000\rangle + \beta|111\rangle \xrightarrow{\mathcal{E}=X_1} |\psi'\rangle = \alpha|100\rangle + \beta|011\rangle \xrightarrow[\Rightarrow G^{\dagger}=X_1]{\delta^{\dagger}=\{-1,+1\}}$

$G^{\dagger}|\psi'\rangle = G^{\dagger}\mathcal{E}|\psi_1\rangle$

$= (X_1)^2 |\psi_1\rangle$

$= |\psi_1\rangle$ ✓

But note that if $\mathcal{E} = X_2 X_3$ (also consistent with $S'$)

then $G = X_1$ still but

$$G\mathcal{E}|\psi_L\rangle = (X_1 X_2 X_3)(\alpha|111\rangle + \beta|000\rangle)$$
$$\neq |\psi_L\rangle \times$$

Terminology

- Parity checks $\Rightarrow$ "stabilizers" (Hermitian operators)
  (eg $\hat{P}_{12}, \hat{P}_{23}$, & $\hat{P}_{13}$)
- "code words" are $+1$ e/states of stabilizers
  eg $|000\rangle$, $|111\rangle$ & $\alpha|000\rangle + \beta|111\rangle$

- "Stabilizers" are generated by a subset "stabilizer generators" $\{\hat{P}_{12}, \hat{P}_{23}\}$
  $\Rightarrow$ Syndrome data is the set of measurement outcomes from the stab. generators
  eg $S' = \{-1, +1\}$

- all stabilizers must commute with each other, & be Hermitian
  $\Rightarrow$ compatible observables
  $\Rightarrow$ simultaneously measurable
  eg $[\hat{P}_{12}, \hat{P}_{23}] = 0$

this example was simple since $P_{12}$ & $P_{23}$ are products of $\mathbb{1}$ & $Z$ operators.

- rules of thumb for commutators:

- Products of $\mathbb{1}$ & $Z$ always commute with other products of $\mathbb{1}$ & $Z$

- also. $\mathbb{1}$'s & $I$'s (or $Y$'s) commute with $\mathbb{1}$'s & $I$'s (or $Y$'s)

- products of $\mathbb{1}$ & $Z$ don't always commute with products of eg $\mathbb{1}$ & $X$

$$[X, Z] = -2iY \quad \text{(check sign)}$$

but $\{X, Z\} = XZ + ZX = 0$

$\Rightarrow$ Products of Paulis $\{\mathbb{1}, X, Y, Z\}$ either commute or anti commute.

- Products of $X$'s commute with products of $Z$'s if they "collide" in even # of places.

eg $[X_1 \otimes X_2, Z_1 \otimes Z_2] = 0$

check $= (X_1 Z_1) \otimes (X_2 Z_2) - (Z_1 X_1) \otimes (Z_2 X_2)$

$= (-iY_1) \otimes (-iY_2) - (iY_1) \otimes (iY_2)$

$= (-i)^2 Y_1 \otimes Y_2 - (+i)^2 Y_1 \otimes Y_2$

$= 0$

- Products anti-commute if # collisions is odd.

(check)

We need to represent logical Pauli operators for logical qubits $\hat{X}_L, \hat{Y}_L, \hat{Z}_L$ s.t

$$\hat{X}_L |0_L\rangle = |1_L\rangle$$
$$\hat{X}_L |1_L\rangle = |0_L\rangle$$

$$\hat{Z}_L |0_L\rangle = |0_L\rangle \quad \text{etc.}$$
$$\hat{Z}_L |1_L\rangle = -|1_L\rangle$$

Then if $\hat{X}_L, \hat{Y}_L$ & $\hat{Z}_L$ should satisfy a Pauli algebra

- find these, eg $|0_L\rangle = |000\rangle$ $|1_L\rangle = |111\rangle$

guess $\hat{Z}_L = Z_1 \otimes \mathbb{1}_2 \otimes \mathbb{1}_3$ or $\mathbb{1} Z \mathbb{1}$ or $\mathbb{1}\mathbb{1}Z$

or $\hat{Z}_L = Z_1 Z_2 Z_3$ (redundancy here)

& $\hat{X}_L = X_1 \otimes X_2 \otimes X_3$ (is unique in this case)

Check that $\hat{Z}_L$ & $\hat{X}_L$ act correctly on the logical code words

take $Z_L = Z_1 Z_2 Z_3$

$X_L = X_1 X_2 X_3$

$\Rightarrow$ calculate $\hat{Y}_L = -i X_L Z_L$ (check sign!!)

$$= -i (X_1 \otimes X_2 \otimes X_3)(Z_1 \otimes Z_2 \otimes Z_3) = (i)(-i)^2 Y_1 \otimes Y_2 \otimes Y_3$$
$$= Y_1 \otimes Y_2 \otimes Y_3$$

- Check they have correct eigenstates

eg $X_L |\pm_L\rangle = \pm |\pm_L\rangle$ $\left(|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)\right)$

guess $|\pm_L\rangle = \frac{1}{\sqrt{2}}\left(|0_L\rangle \pm |1_L\rangle\right)$

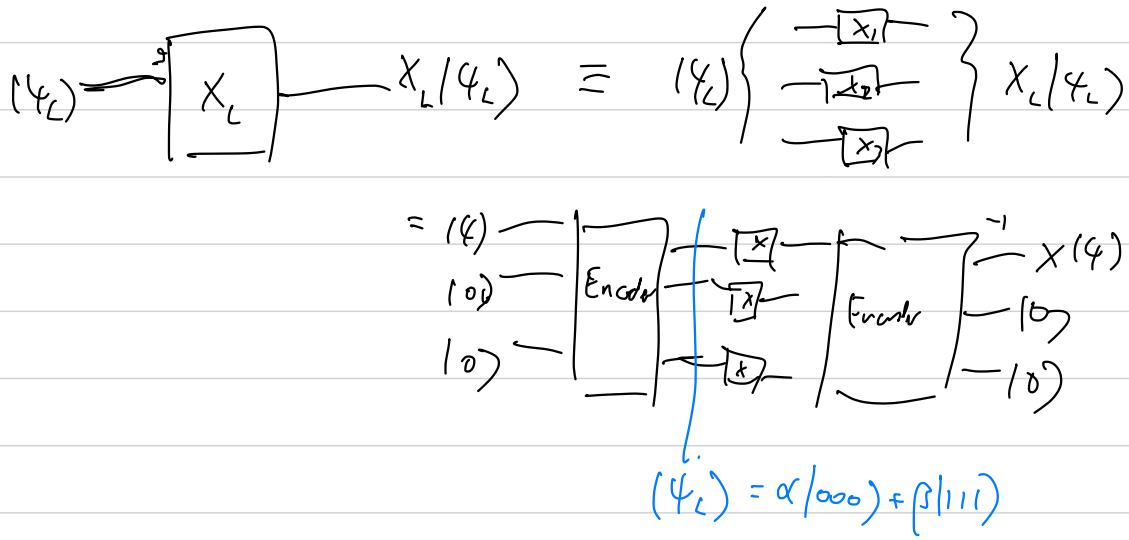
check: $X_L |\pm\rangle_L = \frac{1}{\sqrt{2}}\left(X_L|0_L\rangle \pm X_L|1_L\rangle\right)$

$= \frac{1}{\sqrt{2}}\left(|1_L\rangle \pm |0_L\rangle\right)$

$= \pm\frac{1}{\sqrt{2}}\left(|0_L\rangle \pm |1_L\rangle\right)$

$= \pm |\pm\rangle_L$ ✓

- These logical operators are examples of "transversal" operators:
  - The logical operator is just a product of physical operators acting on each qubit



Logical operators should:

① satisfy their own Pauli algebra

- eg $[X_L, Y_L] = +iZ_L$ (check sign)

$$\{X_L, Z_L\} = 0 = \{X_L, Y_L\} = \{Y_L, Z_L\}$$

② Commute with the stabilisers (generators suffice)

Physically: we need to measure all generators without disrupting code words

Check these eg ① $[X_L, Z_L] = 2iY_L$ (check sign)

$$\{X_L, Z_L\} = (XXX)(ZZZ) + (ZZZ)(XXX)$$

$$= (XZ)\otimes(XZ)\otimes(XZ) + (ZX)\otimes(ZX)\otimes(ZX)$$

$$= 0$$

② Check eg $[X_L, \hat{P}_{12}] = [Z_L, P_{12}] = 0$ + for $\hat{P}_{23}$

$$\{XXX, Z, Z, \mathbb{1}\} = (XZ)\otimes(XZ)\otimes(X\mathbb{1}) - (ZX)\otimes(ZX)\otimes(\mathbb{1}X)$$

$$= (i)^2 Y_1\otimes Y_2\otimes X_3 - (-i)^2 Y_1\otimes Y_2\otimes X_3$$

$$= 0$$

etc

QEC codes consist of:

- Stabiliser operators / stabiliser generators
- code words / code spaces (+1 eigenstates of stabilisers)
- Logical operators which satisfy a Pauli algebra
- stabiliser generator measurement ⇒ syndrome data $\vec{s}$
- $\vec{s} \xrightarrow[\text{decoding}]{\text{error}} C_L(\vec{s})$ is a correction we need to apply
- error decoding is a Bayesian inference calculation

Formalism

"Stabiliser"

⇒ "Lookup table"

| $\vec{s}$ | $C$ |
|---|---|
| $\{p_{12}, p_{23}, \ldots\}$ | $C(p_{12}, p_{23}, \ldots)$ |
| ⋮ | ⋮ |
| ⋮ | ⋮ |

$2^k$ rows

$k$ = # stabiliser generators

but LUT decoding is computationally inefficient in $k$.

The examples of error analysis we looked at were in the "Schrödinger" picture, analysing code words.
⇒ Problem is that the physical state space grows as $2^n$

- The stabiliser formalism just tracks errors & stabiliser generators

$$|\text{stabiliser generators}| = n-k$$

Can do syndrome analysis more efficiently in the "Heisenberg" picture – using algebra of Pauli operators for Pauli errors ("bit flip" $X$, phase flip $Z$, & both $Y$)

eg $|\psi_L\rangle \xrightarrow{\hat{\mathcal{E}}} |\psi'\rangle = \hat{\mathcal{E}}|\psi_L\rangle$ $\quad \mathcal{E} \in$ Pauli group, eg $\hat{X}_j$

measure the stabiliser generators $\hat{S}_i |\psi'\rangle = s_i |\psi'\rangle$
if $|\psi'\rangle$ is an e/state of $\hat{S}_i$

if $\hat{\mathcal{E}}$ & $\hat{S}_i$ are products of Paulis then

we have $|\psi'\rangle = \hat{\mathcal{E}}|\psi_L\rangle \Rightarrow \hat{S}_i|\psi'\rangle = \hat{S}_i \hat{\mathcal{E}}|\psi_L\rangle$

$$= \begin{cases} +\hat{\mathcal{E}}\hat{S}_i|\psi_L\rangle & \text{if } [\hat{\mathcal{E}}, \hat{S}_i] = 0 \\ -\hat{\mathcal{E}}\hat{S}_i|\psi_L\rangle & \text{if } \{\hat{\mathcal{E}}, \hat{S}_i\} = 0 \end{cases}$$

So syndrome bit $s_i$ associated to $\hat{S}_i$ is $\pm 1$

- Measure all stabiliser generators to get syndrome data $S = \{s_1, s_2, \ldots\}$ & this is independent of $|\psi_L\rangle$

⇒ Syndrome data for an error $\hat{\mathcal{E}}$ depends on

the commutations $[\mathcal{E}, \hat{S}_i]$ only, not $|\psi_c\rangle$

eg for the repetition code:

$$\hat{S}_1 = \hat{P}_{12} = Z_1 Z_2 \qquad \hat{S}_2 = P_{23} = Z_2 Z_3$$

| Errors $\mathcal{E}_l$ | commute/anticommute $\hat{S} = \{\hat{S}_1, \hat{S}_2\}$ | $G'(S')$ [look up table decoder] |
|---|---|---|
| $\mathbb{1}$ | $\{+,+\}$ | $\mathbb{1}$ |
| $X_1$ | $\{-,+\}$ | $X_1$ ✓ |
| $X_2$ | $\{-,-\}$ | $X_2$ |
| $X_3$ | $\{+,-\}$ | $X_3$ |
| $X_1 X_2$ | $\{-,-\}$ | $X_2$ ✗ |
| $X_1 X_3$ | $\{-,+\}$ | $X_1$ |
| $X_2 X_3$ | $\{+,+\}$ | $\mathbb{1}$ |
| $X_1 X_2 X_3$ | | |

- Stabilisers & syndrome data can detect any error that anti-commutes with at least 1 stabiliser generator.
- Can't detect (or correct) errors that commute with all stabilizer. (generators)

In the example it can't detect $\mathcal{E} = X_1 X_2 X_3 = X_L$

- So errors that coincide with logical operators are undetectable

Problem 3

3-bit repetition code
- can detect / correct any single bit-flip $\hat{X}_j$ (+ reduce the logical error rate of bit-flips) ✓
- can't detect (or correct) phase errors $\hat{Z}_j$

eg $\hat{Z}_1(\alpha|000\rangle - \beta|111\rangle)$ ∈ code space
is a valid code state

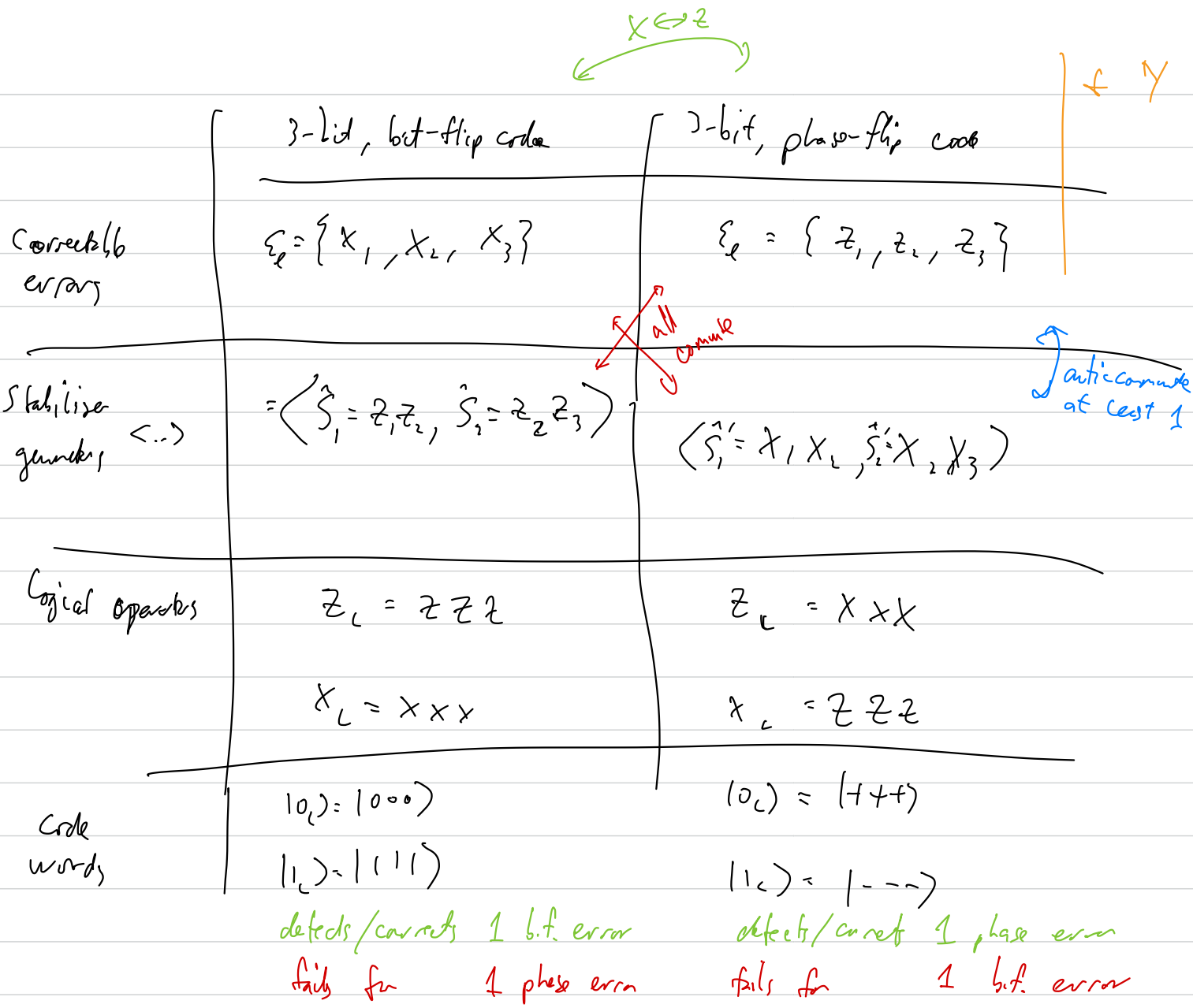
recall that $\hat{Z}_L = \hat{Z}_1$ was a suitable choice for a logical $\hat{Z}$ operator.

- So how do we design a code that can detect + correct any single-qubit Pauli error $\{X_i, Y_i, Z_i\}$

Solution 3a

Codes to detect phase errors

by analogy with the 3-bit rep code




So the 3-bit phase-flip code will (check this)

- detect & correct 1 phase error ✓
- but can't detect b.f. errs ✗

## Question

– is there a quantum code that can correct any single qubit quantum error? $\{X_i, Y_i, Z_i\}_{i=1,2,\ldots,n}$ ($3n$)

## Counting argument

assume we want to protect $k=1$ logical qubit with $n$ physical qubits, $n-k$ stabilizer generators

| $n$ | # possible single qubit errors $= 3n$ | # distinct syndrome bit strings $= 2^{n-k}$ |
|---|---|---|
| 3 | 9 | 4 |
| 4 | 12 | 8 |
| 5 | 15 | 16 |

There is a $n=5$ "qubit code" that can detect & correct any single qubit Pauli error

- $n=5$ physical qubits
  $k=1$ logical qubit
  $n-k=4$ stabilizer generators

find: $\langle \hat{S}_1, \hat{S}_2, \hat{S}_3, \hat{S}_4 \rangle$, $X_L$, $Z_L$

that satisfy suitable commutation & anti-commutation relations

- We can use the 2 3-bit repetition codes (bit flip + phase errors)

Can we combine these to detect any $X$ or $Z$ error. (+ "hopefully" $Y = iXZ$ errors too)

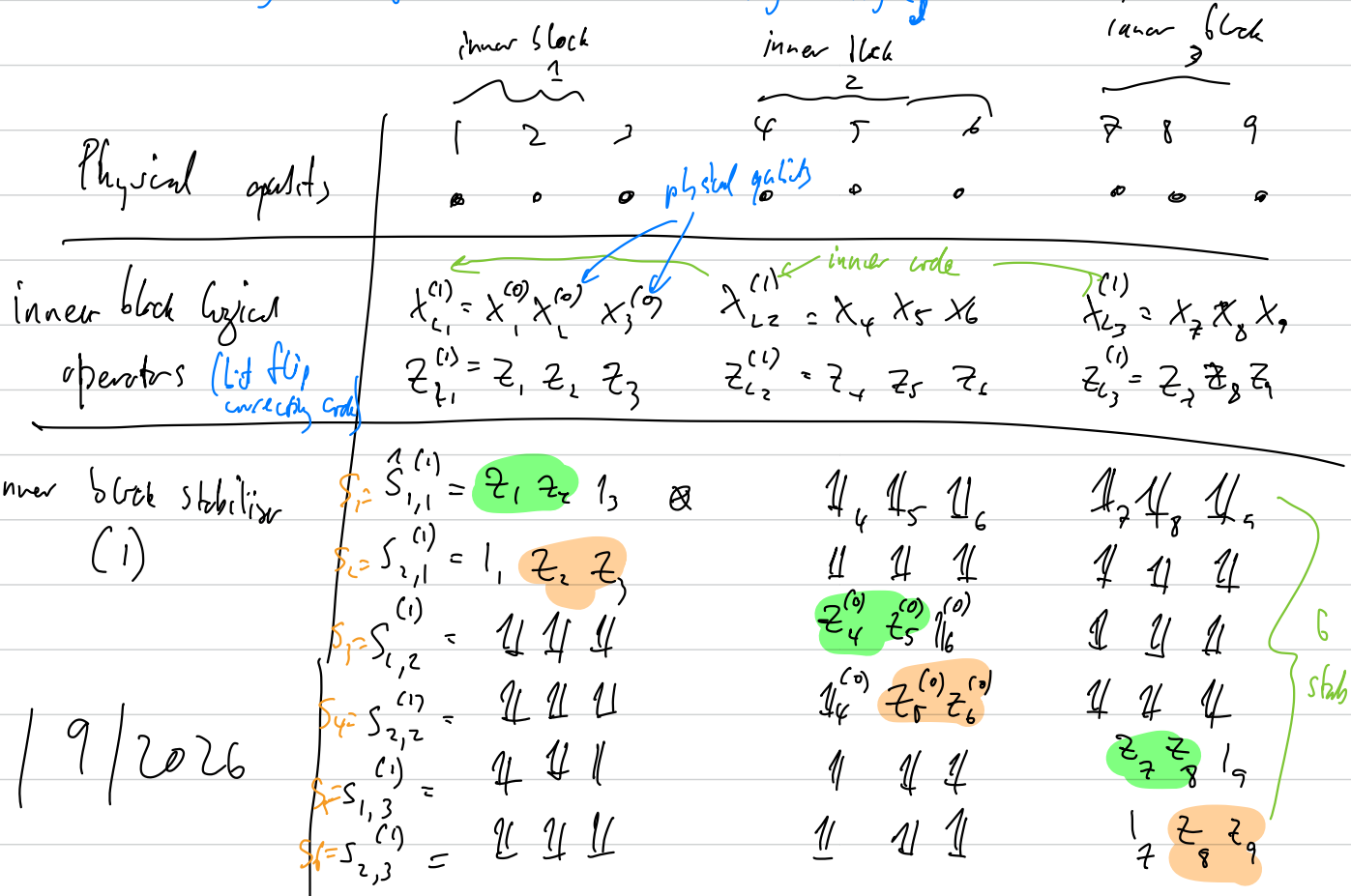
**Solution 3b** — Concatenate the two repetition codes

"Shor code"

- "inner" code to detect/correct 1 b.f error ($\{I, X_j\}$)
- "outer" code " " 1 phase error ($\{I, Z_j\}$)

$\Rightarrow$ by construction also $Y_j = iX_jZ_j$ error.

**Notation**




18/9/2026

$\Rightarrow S_1^{(2)} = X_1 X_2 X_3 \quad X_4 X_5 X_6 \quad \mathbb{1}_7 \mathbb{1}_8 \mathbb{1}_9$

$S_2^{(2)} = \mathbb{1}_1 \mathbb{1}_2 \mathbb{1}_3 \quad X_4 X_5 X_6 \quad X_7 X_8 X_9$

} +2 stab outer code

outer code Logical operators

$X_L^{(2)} = Z_{L_1}^{(1)} \otimes Z_{L_2}^{(1)} \otimes Z_{L_3}^{(1)} = Z_1 Z_2 Z_3 \; Z_4 Z_5 Z_6 \; Z_7 Z_8 Z_9$

$Z_L^{(2)} = X_{L_1}^{(1)} \otimes X_{L_2}^{(1)} \otimes X_{L_3}^{(1)} = X_1 X_2 X_3 \; X_4 X_5 X_6 \; X_7 X_8 X_9$

$\Rightarrow Y_L^{(2)} = (i) X_L^{(2)} Z_L^{(2)}$ (check sign)

$\Rightarrow$ 8 stabilizer generators (6 for inner code + 2 outer)

$n = 9$ qubits

$\Rightarrow$ $k$ = # physical qubits − # stabilizer generators

$= 9 - 8 = 1$ Logical qubit

$\Rightarrow$ Shor code is a $[[9, 1, d=?]]$ code where $d$ is the distance

Claim: the Shor code can detect & correct any single qubit ~~or~~ Pauli error $\{X_1, Y_1, Z_1, X_2, Y_2, Z_2, \ldots Z_9\}$

$|\text{"}| = 27$

# correctable errors $= \left\lfloor \frac{d-1}{2} \right\rfloor \Rightarrow d \geq 3$

Check this claim eg a bit flip on qubit 3 would commute with $S_1^{(1)}, S_2^{(1)}, S_3^{(1)} \ldots$

but anticommute with $S_{2,1}^{(1)} = |ZZ\,|||\,|||$

So the syndrome would be $\{+, -, +, +, \ldots\}$

Can enumerate the syndrome for all single qubit $X$ errors & see they are unique.

Aside check that all stabilizer generators commute

- Clear that $S_1, \ldots, S_6$ all mutually commute since they are all products of $X$ & $\mathbb{1}$
- Clear that $S_7$ & $S_8$ both mutually commute (same reason for $Z$ & $\mathbb{1}$)
- Also any $Z$-like stabilizer in $S_1, \ldots S_6$ "collides" with an $X$-like stabilizer on an even # of sites $\{0, 2\}$

  $\Rightarrow$ commute since $[X_i \otimes X_j, Z_i \otimes Z_j] = 0$
- & the generators are all independt.
- also check that $[S_j^{(2)}, X_i^{(2)}] = 0$ & $[S_j^{(2)}, Z_i^{(2)}] = 0$ } even # of "collisions" of $X$'s & $Z$'s.

Bit-flip errors give unique syndrome bits

$\Rightarrow$ uniquely identify any of $X_1, X_2, \ldots X_9$ errors.

## Subtlety of the Shor code

- The $X$-like stabilizers are "degenerate"

because there are multiple $Z_j$ errors the same syndrome.

eg errors $Z_1$, $Z_2$ or $Z_3$ have syndrome bits

$$S'_{123} = \{+, \underbrace{+ \cdots +}_{S_1 \cdots S_6}, \underbrace{- +}_{S_7\ S_8}\}$$

& $Z_4, Z_5, Z_6 \Rightarrow S'_{456} = \{\underbrace{+ \cdots +}_{S_1\ S_6}, \underbrace{- -}_{S_7\ S_8}\}$

& $Z_7, Z_8$ or $Z_9 \Rightarrow S'_{789} = \{\underbrace{+ \cdots +}_{S_7\ S_8}, +, -\}$

So if we get $S'_{123}$ we don't know which qubit 1, 2 or 3 suffered an phase error

Is this a problem?

eg if $\mathcal{E} = Z_3 \Rightarrow S'_{123} = \{\underbrace{+, \ldots +}_{S_7\ S_8}, - +\}$

we would then have to guess $G(S'_{123}) = Z_1$ or $Z_2$ or $Z_3$?

Let's say $G(S'_{123}) = Z_1$

So $|\psi\rangle \xrightarrow{\text{error}} |\psi'\rangle = \mathcal{E}|\psi_L\rangle \xrightarrow[\text{circuit}]{\text{Syn}} |\psi''\rangle = \mathcal{C}(S_{1,2,3})\,\mathcal{E}|\psi_L\rangle$

($|\psi_L\rangle$: code space)

$= Z_1 \cdot Z_3 |\psi_L\rangle \in$ code space

$= |\psi_L\rangle$

However $Z_1 Z_3 = S_1 S_3$

So $|\psi''\rangle = S_1 S_3 |\psi_L\rangle = S_1((+1)|\psi_L\rangle)$

since $|\psi_L\rangle$ is a $+1$ e/state of all stabilizers

$= (+1)\, S_1|\psi_L\rangle = (+1)^2|\psi_L\rangle = |\psi_L\rangle$

So the $\mathcal{C}(S)\mathcal{E} \in$ Stabilizer group [they are products of the stabilizer generators]

$\Rightarrow$ code space is unchanged by $\mathcal{C}(S)\mathcal{E}$.

So degeneracy for phase error syndromes is OK.

- Check a single qubit $Y_j$ error

$Y_j = (i)\, X_j Z_j$

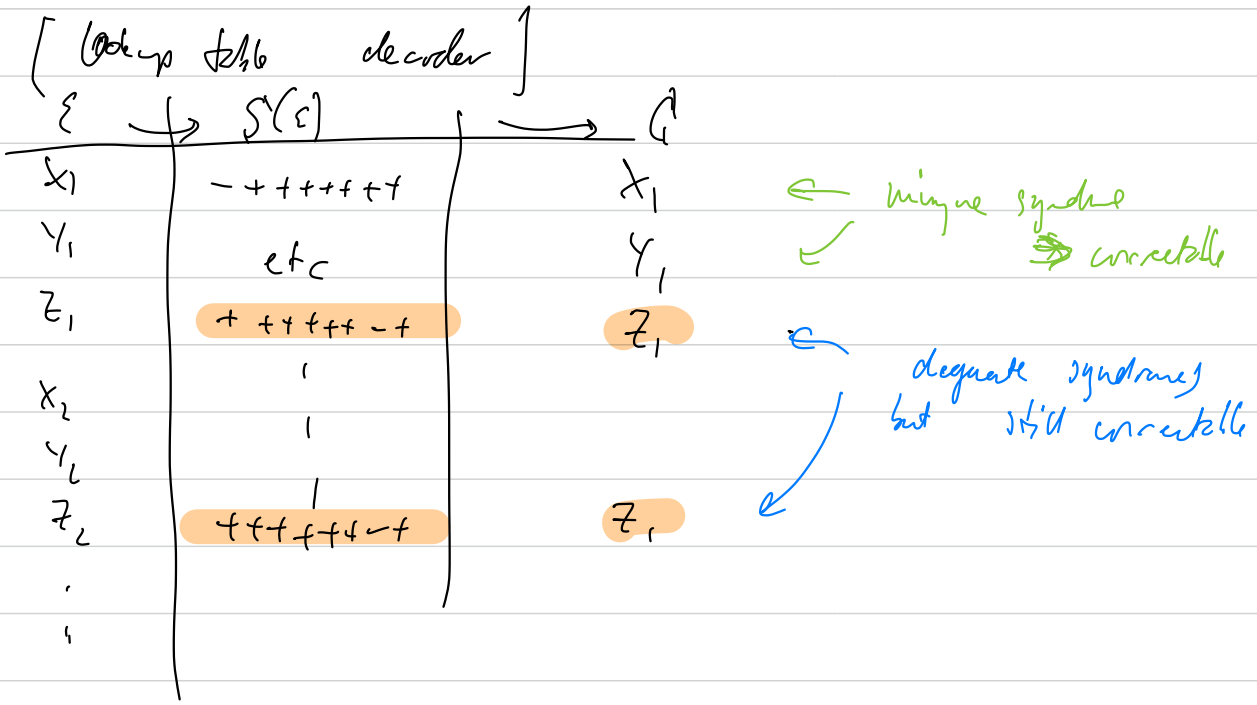
So this will give a mixed syndrome eg

$Y_1 \rightarrow S_{Y_1} = \{\underbrace{-++++}_{S_1, \ldots, S_6}, -, +\ \}$ ($S_7$, $S_8$)

$\Rightarrow$ correct with $G(\bar{S}) = X_1 Z_1 \propto Y_1$

$\Rightarrow$ by exhaustive checking we get correctable syndromes for all 27 single qubit Pauli errors.

[ look up table decoder ]




- $X_L Z_L \propto Y_L = YYY\,YYY\,YYY$ is traversed too
- $\{X_L, Z_L\} = 0$ has the correct logical Pauli algebra.

Note that there are many equivalent presentations of logical operators

- A logical operator acts on logical states

eg $|\psi_L'\rangle = X_L |\psi_L\rangle = X_L (S_j |\psi_L\rangle)$ since $S_j |\psi_L\rangle = (+1)|\psi_L\rangle$

$= (X_L S_j) |\psi_L\rangle$

so $X_L$ & $X_L' = X_L S_j$ act identically on logical states.

eg if $S_5 = S_1 = ZZ1\ 111\ 111$

$X_L = ZZZ\ ZZZ\ ZZ$ has weight 9

then $X_L' = X_L S_1 = 11Z\ ZZZ\ ZZZ$ has weight 7

& $X_L'' = X_L S_1 S_3 = 11Z\ 11Z\ ZZZ$ " " 5

$X_L''' = X_L S_1 S_3 S_5 = 11Z\ 11Z\ 11Z$ " " 3

So the minimum weight logical $X_L$ is $d=3$ up to stabilizer multiplication,

So the Shor code has $d=3$

Shor code is a $[[9, 1, 3]]$ code.

Comments on the Shor code:

- The first concatenated q.e.c that can detect & correct an single qubit Pauli error. ✓
- if the single qubit error rate is $p/27$ per error (so any error happens with prob $p$)

then the prob of an uncorrectable 2-qubit error

$\sim p_{2\text{qubit}} \sim \binom{27}{2} \left(\frac{p}{27}\right)^2 \left(1 - \frac{p}{27}\right)^{9-2}$

$$= \beta p^2$$

where $\beta$ is a combinatorial factor.

So if $\beta p^2 < p$ then the code will improve the logical error rate compared to the physical error rate,

$\Rightarrow$ Threshold for error correction to be $\frac{1}{\beta}$.

General remarks

- We have only shown that the Shor code can correct any 1-qubit Pauli error. $\{X_i, Y_i, Z_i\}$

$\Rightarrow$ QEC Condition $\Rightarrow$ if a code can correct $\{\xi_j\}$

then it can also correct Given combinations of these errors $\{\alpha \xi_1 + \beta \xi_2, \dots\}$

[Nielsen + Chuang]

- Many QEC codes exist today:
  - "Surface code"
  - "colour code"

- holographic codes

are all stabilizer codes that have stabilizer generators + Logical operators

⇒ can correct $> 1$ single qubit error.

- Q.E.C is still being developed

⇒ key to scalable quantum computing